

CO 1 - AT - 1 - Short Answer Test

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Course Code: CSA 0702

Course Name: Computer Networks

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ASSESSMENT TOOL - 1

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DATE: 17/07/26

SHORT ANSWERS:

1. Given:

- * File Size = 150mb
- * Bandwidth = 50 Mbps
- * Segment Size = 1500 bytes
- * Data Link Layer overhead = 40 bytes / frame

Formula:

1. Number of Segments = File Size / Segment Size
2. Total Overhead = Number of Frames $\times$ overhead per Frame
3. Transmission Time = File Size (bits) / Bandwidth.

Calculations:

Convert file size

$$150 \text{ MB} = 150 \times 1,000,000 = 150,000,000 \text{ bytes}$$

Number of Segments

$$\begin{aligned} &= 150,000,000 / 1500 \\ &= 100,000 \text{ Segments} \end{aligned}$$

Since each Segment Become One frame

$$\text{Frame} = 100,000$$

Total Data Link Overhead

$$\begin{aligned} &= 100,000 \times 40 \\ &= 4,000,000 \text{ bytes (4MB)} \end{aligned}$$

Transmission Time

$$150 \text{ MB} = 150 \times 8 = 1200 \text{ Mb}$$

$$\begin{aligned}\text{Time} &= 1200 / 50 \\ &= 24 \text{ seconds}\end{aligned}$$

Final Answer:

- * Number of Segments = 100,000
- * Number of Frames = 100,000
- * Total Overhead = 4,000,000 bytes (4MB)
- * Transmission Time = 24 seconds.

2. Given:

- * Sliding window size = 6 frames
- * Total frames = 48
- * one frame lost in every window.

Formula:

1. Windows = Total Frames / Window Size
2. Retransmissions = One Lost frame $\times$ Number of windows.

Calculations:

$$\begin{aligned}\text{Number of Windows} &= 48 / 6 \\ &= 8 \text{ windows.}\end{aligned}$$

Retransmissions

$$\begin{aligned}&= 8 \times 1 \\ &= 8 \text{ frames}\end{aligned}$$

Final Answer:

- * Number of Windows = 8
- * Total Retransmissions = 8 frames.

Given:

- * IP Packet Size = 12,000 bytes
- * MTU = 1500 bytes
- * IP Header = 20 bytes

Formula:

1. Data per Fragment = MTU - Header
2. Number of Fragments = Packet Size / Data per Fragment
3. Header Overhead = Number of Fragments $\times$ Header Size

Calculations:

Data per Fragment

$$\begin{aligned} &= 1500 - 20 \\ &= 1480 \text{ bytes} \end{aligned}$$

Number of Fragments

$$\begin{aligned} &= 12000 / 1480 \\ &\approx 8.11 \end{aligned}$$

Therefore

9 fragments

Header Overhead

$$\begin{aligned} &= 9 \times 20 \\ &= 180 \text{ bytes} \end{aligned}$$

Final Answer:

- * Number of Fragments = 9
- * Total Header Overhead = 180 bytes

4.

Given:

- * Sliding window Size = 6 frames
- * Total frames = 12
- * One frame Lost per Window

Formula:

1. windows = Total Frames / window Size
2. Retransmissions = one Lost frame $\times$ Number of Windows

Calculation:

$$\begin{aligned}\text{windows} &= 12 / 6 \\ &= 2 \text{ windows}\end{aligned}$$

Retransmissions

$$= 8 \times 1$$

$$= 8 \text{ frames}$$

Final Answer:

- * Number of windows = 2
- * Total Retransmissions = 8 frames

5.

Given:

- * Video File Size = 600 MB
- * Checkpoint every 60 MB
- * Failure after 390 MB

Formula:

1. Checkpoints = Data Transferred before failure / checkpoint interval.
2. Retransmitted Data = Failed Data - Last Checkpoint

Calculation.

Checkpoint before failure

$$390 / 60$$

$$= 6.5$$

fore, Completed checkpoints = 6

✓ Check point =

$$6 \times 60$$

$$= 360 \text{ MB}$$

transmitted Data

$$= 390 - 360$$

$$= 30 \text{ MB}$$

Final Answer :

* Checkpoints Created = 6

* Data Retransmitted = 30 MB

Given.

* Original File Size = 900 MB

* Compression = 40%.

* Encryption Overhead = 5%. Compressed Size

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Formula:

$$1. \text{ Compressed Size} = \text{original Size} \times (100 - \text{Compression \%}) / 100$$

$$2. \text{ Encrypted Size} = \text{compressed Size} \times (1 + \text{overhead \%})$$

$$\text{Percentage Reduction} = \frac{\text{original Size} - \text{Encrypted Size}}{\text{original Size}} \times 100$$

Calculation:

Compressed file Size

$$= 900 \times 60\%$$

$$= 540 \text{ MB}$$

Encryption Overhead

$$= 540 \times 5\%$$

$$= 27 \text{ MB}$$

Encrypted File Size

$$= 540 + 27$$

$$= 567 \text{ MB}$$

Total Percentage Reduction.

$$= ((900 - 567) / 900) \times 100$$
$$= 37\%$$

Final Answer:

- * Compressed File Size = 540 MB
- * Encrypted File Size = 567 MB
- * Total Reduction = 37.1.
- *

7. Given:

- * Data Size = 3 GB
- * Throughput = 120 Mbps
- * Request Size = 3 MB

Formula:

1. Number of Requests = Total Data / Request Size
2. Transmission Time = Total Data (mb) / Throughput

Calculation:

Convert data

$$3 \text{ GB} = 3000 \text{ MB}$$

Number of Requests

$$= 3000 / 3$$

$$= 1000 \text{ request}$$

Convert Data to Mega bits

$$3000 \text{ MB} \times 8$$

$$= 24,000 \text{ Mb}$$

Transmission Time

$$= 24,000 / 120$$

$$= 200 \text{ seconds.}$$

Final Answer:

- * Number of Requests = 1000
- * Transmission Time = 200 seconds.

n:

- * Application Layer Delay = 4ms
- * Transport Layer Delay = 3ms
- * Network Layer Delay = 5ms
- * Data Link Layer Delay = 1ms
- * Physical Layer Delay = 1ms
- * Propagation Delay = 18ms
- * Transmission Delay = 12ms

Formula:

Total Delay = Processing Delay + Propagation Delay + Transmission Delay.

Calculation:

Processing Delay.

$$= 4 + 3 + 5 + 2 + 1$$

$$= 15 \text{ ms}$$

Total Delay.

$$= 15 + 18 + 12$$

$$= 45 \text{ ms}$$

Final Answer:

- * Total Processing Delay = 15 ms
- * Total End to End Delay = 45 ms.

9.

Given:

- * useful Data = 80 MB
- * Frame size = 1600 bytes
- * overhead per frame = 80 bytes.

Calculation:

Payload per frame

$$= 1600 - 80$$

$$= 1520 \text{ bytes}$$

Convert data

$$80 \text{ MB} = 80,000,000 \text{ bytes.}$$

Number of Frames

$$= 80,000,000 / 1520$$

$$\approx 52,631.58$$

Therefore.

52,632 frames

Total Overhead.

$$= 52,632 \times 80$$

$$= 4,210,560 \text{ bytes}$$

Total Data Sent

$$= 80,000,000 + 4,210,560$$

$$= 84,210,560 \text{ bytes}$$

Transmission Efficiency

$$= (80,000,000 / 84,210,560) \times 100$$

$$\approx 95\%$$

Final Answer:

* Number of Frames = 52,632

* Total Overhead = 4,210,560 bytes

* Transmission Efficiency = 95%

Given:

* Original medical record = 240 MB

* Compression = 30%

* Segment Size = 1400 bytes

* Data Link Overhead = 60 bytes / frame

Calculation.

Compressed file size

$$= 240 \times 70\%$$

$$= 168 \text{ MB}$$

to bytes

$$168 \text{ MB} = 168,000,000 \text{ bytes}$$

Number of Segments

$$= 168,000,000 / 1400$$

$$= 120,000 \text{ Segments}$$

Total Overhead..

$$= 120,000 \times 60$$

$$= 7,200,000 \text{ bytes}$$

$$= 7.2 \text{ MB}$$

Final Data Transmitted

$$= 168,000,000 + 7,200,000$$

$$= 175,200,000 \text{ bytes}$$

$$= 175.2 \text{ MB}$$

Final Answer:

* Compressed File Size = 168 MB

* Number of Segments = 120,000

* Total protocol overhead = 7,200,000 bytes

* Final Data Transmitted = 175.2 MB